6 (a) Mass defect is the <u>difference in mass between</u> the mass of the individual / <u>separated protons and neutrons and the (combined) nucleus.</u> [A1]

(b) (i) There are 2 protons and 1 neutron in the nuclide helium−3.

$$\text{mass defect} = 2\times 1.00728u + 1.00867u - 3.014931u$$
$$= 0.008299u$$ [B1]

$$\text{binding energy} = \frac{0.008299\times 1.66\times 10^{-27}\times\left(3\times 10^{8}\right)^{2}}{1.6\times 10^{-19}}$$
$$= 7749191\ \text{eV}\ \left(\text{or } 1.239\times 10^{-12}\ \text{J}\right)$$ [B1]

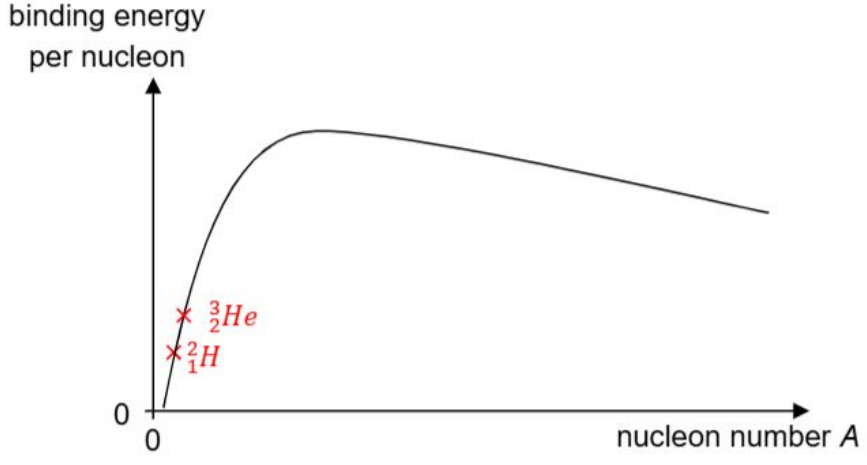
$$\text{binding energy per nucleon} = \frac{7749191\ \text{eV}}{3\times 1\times 10^{6}}$$ [M1]

$$= 2.58\ \text{MeV}$$
$$= 2.6\ \text{MeV}$$ [A0]

B1 for finding the mass defect
B1 for finding the binding energy in J or eV
M1 for finding the binding energy per nucleon in MeV

(ii) Energy released = (binding energy of helium-3) − (binding energy of deuterium)
$= 2.58\times 3 - 1.11\times 2$ [M1]
$= 5.52$ MeV [A1]

(c) (i)



[A1]

A1 for both elements on the curve

(ii) Helium-3 has a <u>higher binding energy per nucleon</u> than deuterium, [B1]

so helium-3 has a <u>higher binding energy/ product is more stable/ higher mass defect</u>. [B1]
Hence energy is released in the reaction.